

A380X OANS RWY AHEAD without an airport map

The RWY AHEAD advisory from the runways of the sim when no Navigraph airport map is loaded

Branch	pr/19-a380x-oans-rwy-ahead-without-map
Aircraft	A380X
Commits	aac868b24 feat(a380x/oans): show RWY AHEAD without an airport map
Base	pr/18-a32nx-raas (uses its NearbyRunwayProvider and AwarenessRunway)
Status	CI checks pass. Opens after PR 18.

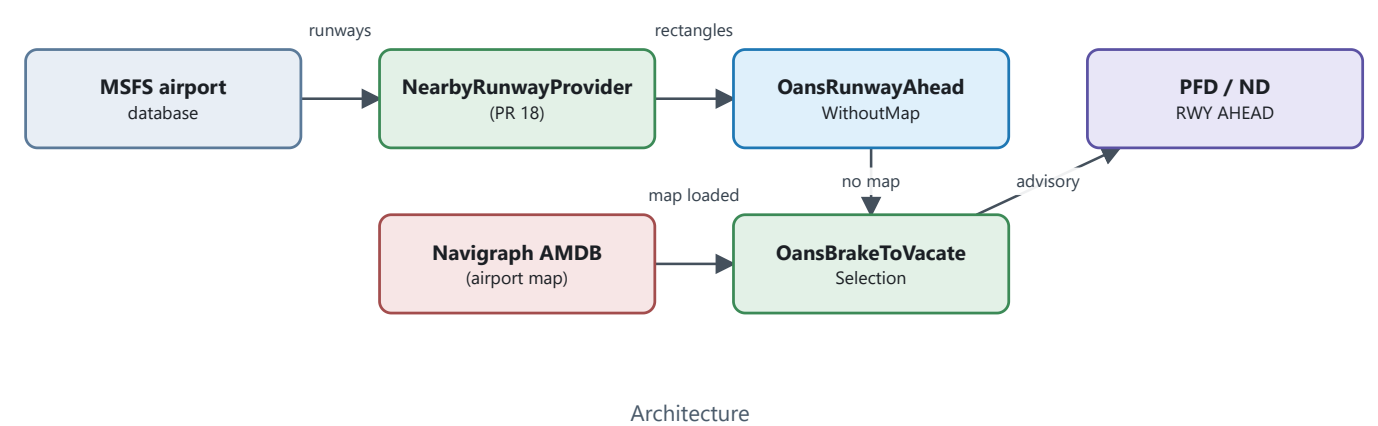
Purpose

The OANS RWY AHEAD advisory needs the Navigraph airport map (AMDB) and stays silent without it, so most users never get it.

Specification

- A380 FCOM DSC-31-20-30-100: ND message RWY AHEAD : CHANGE MODE. This FCOM edition does not describe the detection itself: the rule is the existing FBW OANS RWY AHEAD rule.
- While no airport map is loaded, OANS runs that rule against the runway rectangles of the sim airport database: the volume from the nose to 7 s ahead, 60 m wide, from 1 to 40 kt on the ground, 30 s at most.
- The advisory is issued on the PFD and the ND as with the map. With an airport map loaded, nothing changes.

Component added or changed by the feature MSFS / simulator Aircraft system (systems host, FMS, FWS) Display / page
Data, LVar, storage Crew, flypad, external program



Tests

4 unit tests: `fbw-common/src/systems/instruments/src/OANC/OansRunwayAheadWithoutMap.spec.ts`.

PR description (from the PR draft)

From `pr-drafts/19-a380x-oans-rwy-ahead-without-map.md`, section "Summary of Changes" onwards.

Summary of Changes

The OANS RWY AHEAD advisory needs the Navigraph airport map (AMDB) and stays silent without it, so most users never get it.

While no airport map is loaded, OANS now runs the same rule as with the map (the volume from the nose to 7 s ahead, 60 m wide, from 1 to 40 kt on the ground, 30 s at most) against the runway rectangles of the sim's own airport database, through the shared `NearbyRunwayProvider` (added by #[PR 18 number], the RAAS PR), and issues the advisory on the PFD and ND the same way. With an airport map loaded, nothing changes.

4 unit tests (`fbw-common/src/systems/instruments/src/OANC/OansRunwayAheadWithoutMap.spec.ts`).

Cockpit API Changes

None.

Screenshots (if necessary)

TO ADD: RWY AHEAD on the PFD and ND while taxiing onto a runway without an airport map loaded.

References

- A380 FCOM DSC-31-20-30-100 (ND message RWY AHEAD : CHANGE MODE). This FCOM edition does not describe the detection itself: the rule is FBW's existing OANS RWY AHEAD rule, applied to the sim's runways.

Discord username (if different from GitHub): **TO ADD**

Testing instructions

1. No Navigraph airport map loaded (no subscription, or OANS map not loaded).
2. Taxi slowly towards a runway: RWY AHEAD on the PFD and ND when the runway is within 7 s ahead.
3. Above 40 kt, or airborne: no advisory. With the airport map loaded: the existing behaviour.
4. `npx vitest run fbw-common/src/systems/instruments/src/OANC/OansRunwayAheadWithoutMap.spec.ts` (4 tests).